

THE CIA TRIAD

The CIA Triad is the fundamental model for security.

1. Confidentiality (Protecting Sensitive Information)

Confidentiality ensures that information is only accessible to those authorized to see it.

It follows the **Need-to-Know Principle**.

- **Access Control:** Limiting who can see what.
- **Data Classification:** Labeling data as "Public," "Internal," or "Confidential."
- **Examples:** Employee salary records, customer credit card numbers, and secret product designs.

2. Integrity (Accuracy of Information)

Integrity ensures that information is accurate, complete, and has not been tampered with by unauthorized parties.

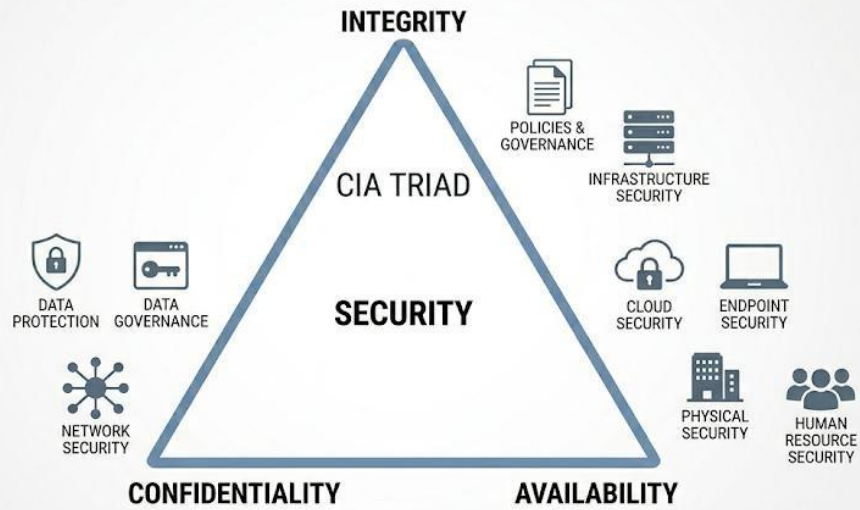
- **Data Validation:** Checking that data entered is correct.
- **Unauthorized Changes:** Preventing a hacker from changing a \$100 bank transfer into a \$10,000 transfer.
- **Examples:** Financial transaction logs, medical prescriptions, and university degrees.

3. Availability (Access When Needed)

Availability ensures that information and systems are ready for use when the business needs them.

- **Business Continuity:** Having a plan to keep working during a disaster.
- **High Availability:** Using redundant systems so if one fails, another takes over.
- **Examples:** An online banking app being accessible 24/7, or a hospital's patient database being online during surgery.

A FOUNDATIONAL SECURITY FRAMEWORK:
A HIERARCHY OF SECURITY DOMAINS: THE CIA TRIAD



The CIA Triad (Confidentiality, Integrity, and Availability) is a foundational model that guides the design, implementation, and evaluation of security protocols across diverse IT domains.